

Template-Based Shot Composition for AI Video Generation

Executive summary

The core idea in your product, a **shot composer layer** that sits between a narrative and downstream image/video generation, is not only sensible but strongly aligned with where the best current evidence points. Across research and tooling, the strongest pattern is **hierarchical decomposition**: convert a broad creative intent into smaller reusable units such as scenes, shots, reference entities, camera instructions, and role-specific constraints; then generate assets per unit; then review and assemble. In other words, the winning abstraction is not “one giant prompt,” but **narrative plan → shot plan → asset generation → review/assembly**. ¹

Industry practice points in the same direction. Google Flow explicitly offers tooling such as **Storyboard Studio**, described as a way to “write a script, create the cast, and visualize a storyboard,” while Flow itself is organized around **Plan, Create, Refine**. Runway frames Gen-4 as a storytelling system where users can “cast your characters, scout your locations, block your scenes,” and preserve world consistency across scenes through references. TikTok’s official creative guidance recommends a reusable **hook-body-close** narrative structure and even calls trends “storytelling templates.” Shopify’s commerce guidance similarly recommends shot lists that include hero images, angles, close-ups, lifestyle context, and size comparisons. These are all, in practice, domain-specific narrative and shot templates. ²

Academic work has moved beyond single-prompt generation toward explicit **multi-scene scripts, storyboards, reference-conditioned continuity, and shot-level control**. VideoStudio uses an LLM to expand a prompt into a multi-scene script with event descriptions, entities, and camera movement. TeViS formalizes text-synopsis-to-storyboard retrieval. DrawVideo, DreamShot, and StoryBlender all treat storyboarded or shot-decomposed generation as a first-class problem for long-form coherence and controllability. Direct2V and FlowZero use LLMs as frame- or scene-level directors. This is highly relevant to your hypothesis: the field increasingly treats a mid-level plan as necessary for coherence and control. ³

Your own uploaded materials already embody this architecture. The PRD defines a reel as

`1 script → N shots → N images → N clips → 1 reel`, adds a Shot node, separates descriptive shot controls from API-native generation controls, and places versioning plus designer review around each AI-assisted step. The sample script file already uses a rich, repeatable structure: hook, strategic objective, AI production type, shot-by-shot visual script, on-screen text, voiceover, sound design, caption, CTA, thumbnail hook, and QC notes. In other words, your product intuition is not just plausible, it is already partially operationalized. [filecite\turn0file1](#) [filecite\turn0file0](#)

My overall conclusion is that **yes, template- or narrative-based shot composition is extendable beyond D2C**, but only if you treat templates as **domain-constrained communicative jobs**, not rigid scripts. The generalized architecture should therefore be domain-agnostic at the platform level, while the **template library, compliance logic, performance priors, and evaluation rules** remain domain-specific. This is a

strong fit for commerce, education, training, real estate, gaming, and some news formats; it becomes harder in domains with strict legal balance requirements or highly open-ended artistic goals. ⁴

Definitions and taxonomy

For this report, a **shot template** is a reusable structural unit that specifies the communicative job of a shot, the kind of visual evidence it should contain, and the constraints under which it should be generated. A **shot idea** is a concrete realization of that template for one campaign or concept. A **shot node** is the system object that stores one shot's intent, references, constraints, promptable description, and output history. A **narrative template** is the ordered set of shot templates for a given outcome, such as a D2C reel, a lesson explainer, or a training clip. This vocabulary matches both your uploaded PRD's Shot-node model and current research that decomposes long video into smaller scene or shot units before generation.

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Your production jobs can be organized into a useful hierarchy. At the **narrative level**, the sequence often runs from attention capture to explanation to persuasion to action. TikTok's official model is the clearest public version of this: **hook, body, close**, with value introduced early, product shown in the middle, and CTA at the end. At the **commerce shot level**, Shopify's guidance maps closely to hero, detail, lifestyle, group, packaging, size-comparison, and angle coverage. Combining the two yields a practical D2C ontology that fits your existing categories: **hook, texture/detail, application/demo, hero/product beauty, proof, tutorial/education, lifestyle/context**, and **closure/CTA**. ⁶

A useful product distinction is between **communicative job** and **visual realization**. For example, "proof" may be realized as testimonial text, expert voiceover, ingredient close-up, creator demo, or before-context-without-claims. "Application" may be hand interaction, product-on-skin macro, routine sequence, or step-by-step overlay. "Hero" may be a clean beauty still, a packshot with motion, or a stylized brand tableau. Keeping the job separate from execution is what allows a template system to generalize across domains and brands without collapsing into repetitive aesthetics. TikTok's and Shopify's guidance both implicitly support this separation by recommending reusable structural functions while leaving execution flexible.

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Your sample scripts demonstrate a more advanced taxonomy than many public tools expose. Each reel includes not just shots, but also a **strategic objective, production type, on-screen text structure, voiceover mode, sound design, caption, thumbnail hook**, and **QC notes**. That is exactly the kind of intermediate representation that a serious shot composer should use. It means the system is not just generating visuals; it is generating a **creative plan** with multiple synchronized modalities and review checkpoints. filecite turn0file0

Survey of evidence across research, platforms, and practice

The academic literature now clearly recognizes that longer or more controlled video generation benefits from an explicit planning layer. **StoryGAN** is an older foundational paper that frames "story visualization" as generating a sequence of images conditioned on a multi-sentence story while preserving global consistency. **TeViS** formalizes translation from a text synopsis into an ordered video storyboard. These papers matter because they show that the storyboard or shot-sequence problem is not a fringe UI preference; it is a real modeling problem with its own datasets and benchmarks. ⁸

Recent work is even closer to your product idea. **VideoStudio** converts an input prompt into a **multi-scene script** using an LLM, where each scene includes the event, entities, background, and camera movement, then generates each scene while maintaining cross-scene consistency through references. **DrawVideo** decomposes long video into controllable shots, each defined by a sketch, appearance prompt, and motion prompt. **DreamShot** and **StoryBlender** explicitly target multi-shot storyboard generation with consistency and editability. **From Shots to Stories** proposes **L-Storyboard** as an intermediate language representation for video editing tasks. Together, these papers strongly corroborate a product architecture in which a shot composer sits between a narrative and downstream generation. ⁹

Research on model architecture also supports factorization. **Emu Video** explicitly splits generation into two stages: first an image conditioned on text, then a video conditioned on the text and generated image. This is powerful evidence for an image-first, motion-second production flow. **Direct2V** uses LLMs as “frame-level directors,” while **FlowZero** introduces an LLM-driven dynamic scene syntax. These works suggest that the most controllable systems benefit from a separable planning representation rather than relying only on raw prompt text. ¹⁰

Official platform materials show the same trend. OpenAI’s Sora research page says the model can generate “multiple shots within a single generated video” while preserving characters and visual style, which is evidence that multi-shot narrative coherence is a recognized capability at the model layer. Google DeepMind’s Veo prompt guide tells users to think in terms of **shot framing and motion, style, lighting, character descriptions, location, action, and dialogue**, which is effectively a shot schema. Google Flow makes the scaffold even more explicit with **Plan / Create / Refine** capabilities and a tool called **Storyboard Studio** that lets a user “write a script, create the cast, and visualize a storyboard.” ¹¹

Runway is one of the clearest product-level examples of a narrative-aware composition layer. Its Gen-4 release emphasizes **world consistency** across characters, locations, and objects; its “Behind the Scenes” page says users can “cast your characters, scout your locations, block your scenes,” and turn a few reference images into “endless characters, scenes and points of view.” That is almost a literal productization of reusable narrative blocks plus asset continuity. ¹²

Outside generative-model vendors, ad and commerce platforms provide the strongest evidence for **domain-specific narrative templates**. TikTok’s official Creative Codes recommend TikTok-first content, trends as “storytelling templates,” and a reusable **hook-body-close** structure with a strong first few seconds and explicit CTA. TikTok’s D2C playbooks further recommend a minimum number of unique creatives, emphasize hook quality, and encourage AI-assisted ideation and script generation through Symphony. Shopify’s photography guidance recommends a shot list that includes hero image, product angles, close-ups, lifestyle context, group shots, and size comparisons. These are effectively pre-generative shot libraries for commerce. ¹³

A noteworthy newer tooling example is **LTX Studio**, whose official site positions the product as an AI platform “for video production” with explicit support for **Script-to-Video**, an **AI Storyboard Generator**, a **Dynamic Storyboard**, a **Timeline Editor**, and controls for camera, keyframes, references, styles, and project-level consistency. It also allows users to start “From Script,” “From a Concept,” “From an Image,” or “From a Video.” That is perhaps the clearest current commercial expression of a general-purpose shot-composer layer. ¹⁴

The patent signal is weaker than the research and platform signal, but it still shows precedent for templated composition. An older example is the patented MovieRide FX “digital cinematographic template” system, which inserted user footage into pre-authored movie templates. It is not modern generative AI, but it demonstrates that templated, reusable cinematic blocks long predate diffusion video and remain commercially meaningful. ¹⁵

Concrete examples and what they imply

Several publicly accessible sources use language that maps almost directly onto your product concept. TikTok says trends can function as **“storytelling templates”**, which is unusually explicit language from a major ads platform. That matters because it validates the notion that patterns in short-form video can be abstracted into reusable narrative forms without losing creative flexibility. ¹⁶

Runway’s own phrase that users can **“cast your characters, scout your locations, block your scenes”** is even more directly aligned with a shot-composer architecture. “Cast” maps to identity/reference objects, “scout” to scene libraries and environments, and “block” to shot planning and sequence composition. The follow-on references system then provides continuity across those units. ¹⁷

Google Flow’s **Storyboard Studio** is another clear signal. The description, **“Write a script, create the cast, and visualize a storyboard,”** is effectively a commercial blueprint for the same abstraction: narrative authoring, entity definition, and shot visualization as distinct steps before final generation. Veo’s prompt guide then breaks a shot down into framing, motion, style, lighting, character, location, action, and dialogue, which can be treated as structured slots in a shot object rather than left trapped inside one long paragraph prompt. ¹⁸

OpenAI’s public Sora research materials are less explicit at the UI-layer, but still important. The official page says Sora can create **“multiple shots within a single generated video”** and persist characters and style across them. That means multi-shot structure is recognized at the model capability layer. Since the Sora app was later discontinued, the strongest current public OpenAI evidence is therefore research-facing rather than product-workflow-facing. ¹⁹

Your own uploaded assets show the same principle already working in a domain-specific context. The sample scripts file moves beyond generic prompts and into a composable creative spec: one reel includes five explicit shots, each with duration, physical staging, light, motion, and a final still hold, while also coordinating intro/body/outro text, sound design, caption, CTA, and QC constraints. The PRD then operationalizes this into nodes: Script, Shot, Prompt, Image Gen, Video Prompt, Video Gen, Brand KB, versioning, and review. That is already a serious intermediate representation, not just a prompt wrapper.

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A generalized architecture for a domain-agnostic shot composer

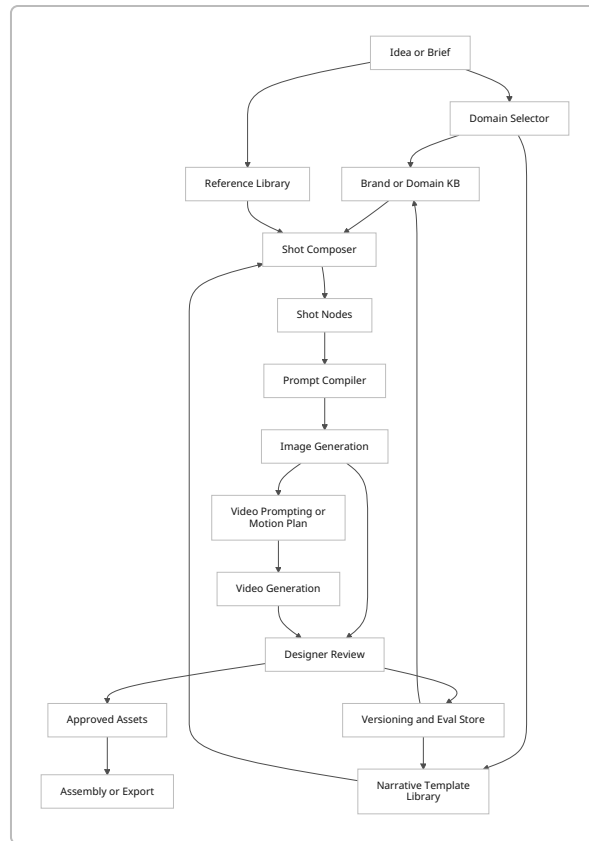
The simplest way to think about the generalized architecture is as a **mid-level planning system**. It should sit between an idea or brief and the underlying generation model. Its job is to turn unstructured intent into a set of structured, versioned, reviewable objects that are easier to generate, evaluate, edit, and reuse. This is consistent with academic systems that use scripts, scenes, or storyboard structures, and with industry tools that separate planning, references, and generation. ²⁰

A robust data model should include at least these entities: **Domain Template, Narrative Template, Shot Template, Shot Node, Reference Asset, Brand or Domain KB, Generation Attempt, Review Decision, and Performance Result**. References and KB objects should be reusable across campaigns. Shot nodes should inherit defaults from templates, then fork into campaign-specific instances. Generation attempts should snapshot prompt, parameters, model, references, and outputs so that learning is attached to the attempt rather than lost in chat history. This is highly consistent with your PRD’s single-source output, shot fan-out, active-version model, and designer review loop. [filecite turn0file1](#)

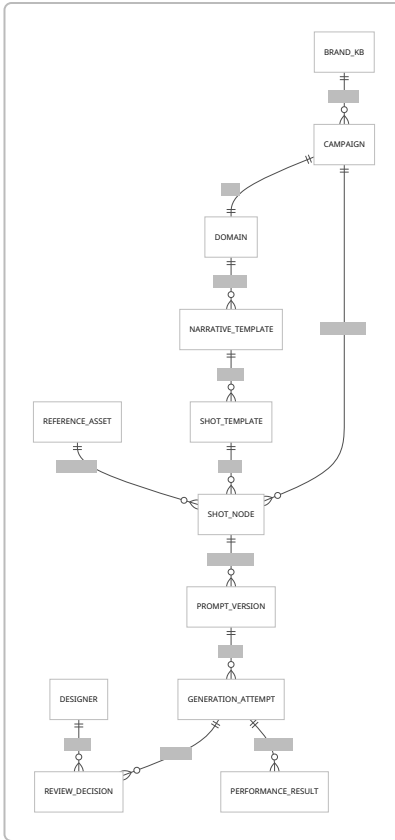
A practical component model looks like this:

Component	Role	Why it matters
Domain Template Library	Stores reusable narrative patterns per domain	Lets the product generalize while remaining constrained
Shot Composer	Expands a narrative into ordered shot nodes	Central planning layer
Brand or Domain KB	Provides voice, claims, compliance, audience, product facts	Prevents off-brand or unsafe outputs
Reference Library	Characters, products, locations, textures, competitor examples, previous best shots	Improves continuity and specificity
Prompt Compiler	Converts structured shot objects into model-specific prompts and controls	Keeps model swaps possible
Generator Layer	Image-first and/or video-first generation	Executes the plan
Versioning and Evals	Stores attempts, edits, outcomes, approval, performance	Creates the learning flywheel
Designer Review	Human edits, rejects, approves, forks	Prevents collapse into low-quality automation

This leads to the following high-level flow:



An ER-style model can be expressed as follows:



The pattern comparison below shows where the strongest public evidence sits:

Platform or source	Explicit reusable unit	Continuity mechanism	Strength of “shot-composer layer” evidence	Relevance to your product
OpenAI Sora research	Multi-shot video capability	Persistent character/style across shots	Moderate at model level, weaker at current product level	Confirms multi-shot structure matters ¹⁹
Google Veo + Flow	Shot schema, storyboard, cast, planning phases	Prompt slots, storyboards, Flow project state	Very strong	Strong blueprint for plan/create/refine and storyboarding ²¹
Runway Gen-4	Characters, scenes, POVs, references	World consistency via references	Very strong	Strong evidence for reusable shot blocks + continuity ¹²

Platform or source	Explicit reusable unit	Continuity mechanism	Strength of “shot-composer layer” evidence	Relevance to your product
Meta research	Multi-modal factorization, image-conditioned video, story visualization	Image-first conditioning and cross-scene consistency	Strong in research, weaker in public workflow tooling	Supports image-first then video pattern ²²
TikTok Ads and Symphony	Hook/body/close, trends, creative quantity	Platform-native hooks, iterative creative variation	Very strong for D2C narrative jobs	Best source for D2C sequence design and scaling rules ²³
Shopify guidance	Hero, angle, close-up, lifestyle, group, size	Shot list and workflow discipline	Strong for commerce shot taxonomy	Best source for commerce visual coverage types ²⁴
LTX Studio	Script-to-video, storyboard generator, dynamic storyboard, timeline	Project-level asset, keyframe, and reference controls	Very strong	Closest current commercial analogue to a generalized shot composer ¹⁴

In the sources reviewed, public primary-source evidence was much stronger for Google Flow, Runway, TikTok, Shopify, OpenAI research, Meta research, and LTX than for Synthesia or Pika on this specific “reusable shot block / narrative composer” abstraction. That does not mean those products lack templates; it means the clearest accessible evidence for your precise architecture was concentrated elsewhere.

Evaluation, risks, and best practices

The benefits of this architecture are substantial. It improves **time-to-first-usable-shot** because designers are no longer inventing the structure from scratch. It improves **variation quality** because each template can generate multiple shot ideas rather than a single thin line item. It improves **consistency** because references, brand guidance, and QC are attached to structured objects. And it improves organizational learning because the unit of analysis becomes the shot and sequence, not a disappearing chat transcript. This is exactly the kind of eval flywheel your PRD is aiming for through versioning, active outputs, and reviewable attempts. ²⁵

The main limitations are also clear. First, templates can lead to **aesthetic collapse** if the system overfits to a small set of visual priors. Second, templates can become **too generic** if they are not grounded in domain jobs and performance data. Third, generative models still struggle with physics, causality, and precise camera trajectories, which means a shot composer can improve control but not eliminate model failure. Fourth, in regulated domains the template layer can accidentally operationalize unsafe claims very efficiently unless compliance is built into the KB, the prompt compiler, and the review step. ²⁶

The highest-risk areas for your use case are legal and brand-related. D2C and wellness content must guard against **implied medical claims**, before/after implications, testimonial exaggeration, and unsupported proof language. Your uploaded scripts already reflect this by softening claims and adding disclaimer

patterns. More broadly, model hallucination, invisible brand drift, over-branding, rights issues in reference assets, and unsafe creator or testimonial mimicry all need formal controls. OpenAI, TikTok, and platform safety pages show that these vendors treat moderation, provenance, and restricted content as core issues, not edge cases. [filecite turn0file0](#) 27

For measurement, the best KPI set is operational plus creative plus business. A workable dashboard would include:

Metric	Definition	Why it matters
Time to first usable shot	Minutes from brief or script to first designer-approved still or clip	Primary productivity metric
First-pass approval rate	Share of generated shots approved without major rewrite	Measures quality of templates and compiler
Variation yield per campaign	Number of distinct acceptable shots produced per template	Measures ideation power, not just generation count
Template reuse rate	Share of campaigns using an existing narrative template	Measures product leverage
Edit distance from generated output	Human rewrite intensity on shot description or prompt	Measures how much the system still misses
Brand-safety failure rate	Attempts rejected for unsupported claims, tone, or identity issues	Critical for trust and legal safety
Performance lift by template	CTR, thumb-stop, CVR, watch time, recall, or downstream action by sequence type	Connects creative structure to business value

These metrics are consistent with TikTok’s emphasis on hook effectiveness, creative quantity, and CTA quality, and with your PRD’s emphasis on versioning, approval, and learning from every attempt. [filecite turn0file1](#) 28

Best practice follows naturally from the evidence. Prompting should be **structured, not poetic by default**: subject, action, setting, framing, lighting, motion, audio, and constraints should each have their own fields. Template design should focus on **communicative jobs**, not just camera labels. Reference usage should distinguish between **identity references** such as product, character, or brand packshot, and **style references** such as mood, film stock, or pacing. Human review should happen at three moments: after narrative expansion, after still generation, and before final video export. The MVP should start with one or two domains and a small library of proven templates, then add automatic performance-based ranking only after enough reviewed data is collected. 29

A sensible MVP roadmap is:

Phase	Scope	Deliverable
Foundation	D2C only, manual template selection, shot-node editing, image-first workflow	Working shot composer for commerce reels
Guided generation	Brand KB slices, reference library, prompt compiler, multi-idea generation per shot type	Better first-draft shot ideas
Review and learning	Versioning, approval states, failure labeling, template analytics	Eval flywheel begins
Video expansion	Motion prompts, start-frame conditioning, sequence continuity checks	End-to-end reel asset workflow
Domain expansion	Education, real estate, training, gaming, regulated pilots	Domain-agnostic platform with domain libraries

Generalization beyond D2C

The principle generalizes well whenever a domain has repeatable **communicative jobs**, reasonably stable **proof formats**, and reusable **constraints**. D2C is simply the obvious first wedge because its shot grammar is already highly legible: hook, show product, show benefit, show context, show proof, ask for action. But the same abstraction appears in many other short-form domains. The key is to generalize the **architecture**, not the exact shot list. ³⁰

A cross-domain template table makes this clearer:

Domain	Example template A	Example template B	Example template C	Key constraint difference
D2C commerce	Hook → texture → application → hero → CTA	Problem → demo → proof → offer	Lifestyle → ingredient story → packshot	Performance and claims must both be optimized
Education	Curiosity hook → concept visual → worked example → recap	Misconception → correction → miniature proof	Topic intro → step-by-step board → takeaway	Accuracy and pedagogical sequencing matter more than brand style
News explainer	Headline → context → evidence visual → implication	What happened → why it matters → what changes now	Myth vs fact → source-backed explanation	Verification, source attribution, and neutrality are central
Real estate	Hook exterior → layout reveal → hero room → neighborhood cue	Price/value anchor → amenity proof → lifestyle vignette	Before context → renovation detail → CTA	Spatial truthfulness and non-misleading representation matter

Domain	Example template A	Example template B	Example template C	Key constraint difference
Gaming	Visual hook → mechanic reveal → payoff moment	Character intro → world reveal → challenge → reward	Patch note → gameplay proof → CTA	Fast pacing, effects, and clear mechanic visibility matter
Enterprise training	Objective → scenario → correct action → recap	Mistake → consequence → better practice	Policy point → example → check-for-understanding	Compliance, clarity, and accessibility dominate
Pharma or healthcare marketing	Symptom context → product mechanism framing → fair-balance close	HCP explainer → evidence visual → indication/ISI	Patient support flow → usage instructions → safety	Highest legal and regulatory burden; script and claims review are mandatory

For each of these domains, the reusable unit is still a narrative-template-backed shot sequence, but the validation criteria differ. In education, the right question is whether the concept was understood. In news, it is whether the sequence is balanced and sourced. In real estate, it is whether the property is represented accurately. In training, it is whether the viewer can perform the required task. In pharma, it is whether every generated sequence respects approved claims, labeling, and fair-balance obligations. This is why the architecture should be general, while the KB, constraints, and evals remain domain-specific. ³¹

For your product specifically, the strongest generalization path is to define a universal schema such as:

Goal → Audience → Narrative template → Shot jobs → Evidence needed → Compliance rules → References → Prompt slots → Review criteria → Performance metrics

That schema can serve D2C, education, training, real estate, and gaming immediately. News and pharma are still possible, but should be treated as stricter governance layers rather than just new prompt presets. ³²

Sources and limitations

The highest-priority sources for this report were official or primary materials: OpenAI's Sora research page and help article, Google DeepMind's Veo and prompt guide, Google Flow's product page, Runway's Gen-4 release materials, TikTok's Creative Codes and D2C/TikTok Shop playbooks, Shopify's ecommerce photography and workflow guides, LTX Studio's official product site, and recent academic papers on multi-shot, storyboarded, or reference-conditioned generation. ³³

Your uploaded PRD and sample script bundle were also important because they show that your current thinking already matches the structure suggested by the external evidence: script parsing, shot fan-out, structured shot descriptions, Brand KB slices, image-first then motion prompting, versioning, and human review. ³⁴

Two limitations are worth stating plainly. First, in the sources reviewed, I found much richer public evidence for Google Flow, Runway, TikTok, Shopify, OpenAI research, Meta research, and LTX than for Synthesia or Pika on the specific question of explicit, reusable **shot-composer** abstractions. Second, some of the most interesting product functionality in the market is surfaced in apps and demos more than in stable documentation, so the public record is uneven. Even with those caveats, the evidence is strong enough to support the main conclusion: **the principle of narrative- and template-based shot composition is real, useful, and generalizable, and your product direction is well aligned with both current research and the strongest current tooling patterns.** ³⁴

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https://arxiv.org/abs/2401.01256?utm_source=chatgpt.com

² ¹⁸ ³⁴ Google Flow - AI Creative Studio for Video, Images & Custom Tools
<https://labs.google/fx/tools/flow>

⁴ ⁶ ⁷ ¹³ ¹⁶ ²³ ³⁰ Creative Best Practices for TikTok Ads | TikTok For Business Blog
<https://ads.tiktok.com/business/en/blog/creative-best-practices-top-performing-ads>

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<https://openai.com/index/sora/>

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<https://runwayml.com/research/introducing-runway-gen-4/>

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<https://ltx.studio/>

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<https://runwayml.com/gen-4-bts>

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